DISPERSIVE POWER OF GRATING

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Aim of the Experiment

To determine the wave length of the prominent lines of a mercury source by a plane transmission diffraction grating and to calculate the resolving power and dispersive power.

Apparatus Required

(i) Spectrometer
(ii) Plane transmission diffraction grating
(iii) Mercury lamp
(iv) Spirit level

Fig-1: Experimental Setup of grating spectrometer

Theory

Fig-1 shows the complete experimental set up consisting of source and the spectrometer mounted with grating. A diffraction grating is an optically plane glass plate with a number of equidistant, parallel, straight lines ruled on its surface shown as in Fig-2. These lines act as slits of width comparable to the wavelength of visible light. So a diffraction grating consists of 'n' number of slits where the value of n can be changed according to the requirement. In laboratory we use replicas of it by depositing a thin film of gelatine on the original grating. By removing it we paste it on an ordinary glass plate to have the ordinary grating.

When a parallel beam of light from the Mercury source is incident normally on the plane transmission diffraction grating, it gets diffracted through an angle θ from their normal path. Depending on the wavelength, lights of different colour are focused at different points on the screen making the prominent lines at different angles of dispersion. From the Fig-3 it is seen that the path difference between any consecutive rays is given by $(a+b)\ \mathrm{Sin}\theta$.

The point P will be bright if this path difference is an even multiple of $\lambda/2$

i.e. $\mathbf{(a+b)\ \sin\theta = 2n\lambda/2}$

Or $\mathbf{(a+b)\ \sin\theta = n\ \lambda}$

where a is the width of the transparent line and b is the width of the opaque line of the grating as shown in Fig-2, n is the order of the maximum intensity and θ is the angle of diffraction or the angle of dispersion for a particular wave length λ. The value "(a+b)" is a constant for a particular grating and is called the grating constant or grating element. For example, a grating with 15000 lines per inch the value of (a+b) is 2.54/15000 cm. Hence knowing (a+b), n and θ the value of λ can be calculated

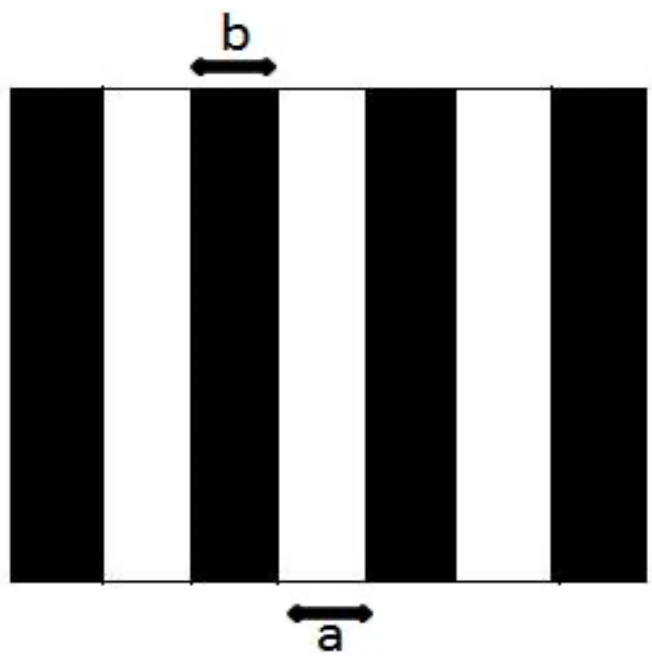

Fig-2: Typical grating

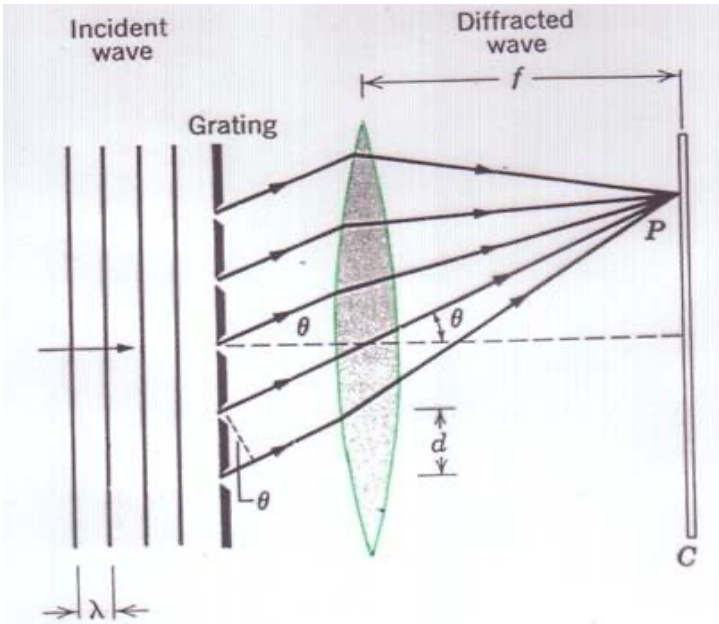

Fig-3: Diffraction through grating

<u>Dispersive Power of Grating</u>

Dispersive power of the grating is given by $d\theta/d\lambda$

We have $(a+b)\sin\theta=n\lambda$

Differentiating both the sides we obtain,

$$d\theta/d\lambda=\frac{n}{(a+b)\cos\theta} \quad \text{or} \quad d\theta/d\lambda=\frac{nN}{\cos\theta}$$

where n is the order of the fringes, N is the number of lines per inch of the grating and θ is the angle of dispersion for the particular order and particular colour.

Thus after knowing n, N and θ the dispersive power of the grating $d\theta/d\lambda$ *can be calculated.*

<u>Resolving Power (R.P.) of Grating</u>

The Resolving power of grating is defined as the ratio of the wave length of any spectral line to the difference of wave length between this line and a neighbouring line such that the two lines appear to be just resolved. Thus the resolving power of the grating is given by

$$R.P = \lambda/d\lambda$$

As per Rayleigh's criteria, the two lines will be exactly resolved if the principal maxima of one line falls on the secondary minima of the other line on either side of its principal maxima. The direction of the nth primary maximum for a wave length λ is given by $(a+b) \sin \theta = n\lambda$. Similarly the direction of the primary maximum for the wave length $\lambda+d\lambda$ is given by

$$(a+b) \sin (\theta+d\theta) = n (\lambda+d\lambda).$$

The two lines will be just resolved, if the angle of diffraction $(\theta+d\theta)$ also corresponds to the direction of the secondary minimum after the nth primary maximum for wave length λ. This is possible if the extra path difference introduced is λ/N, where N is the number of lines per inch of the grating. So we *have* $(a+b) \sin (\theta+d\theta) = n\lambda + \frac{\lambda}{N}$

Hence $\underline{n (\lambda+d\lambda) = n\lambda + \frac{\lambda}{N}}$ *or* $\underline{\lambda/d\lambda=Nn}$

This measures the resolving power of the grating

Procedure

(a) Adjustment of the Spectrometer:

(i) Level the spectrometer properly with the help of a spirit level by levelling the collimator, telescope and prism table etc.

(ii) Switch on the mercury source and focus the image of the slit in the telescope. With the help of a prism adjust the spectrometer for parallel rays by Schuster's method as described in Appendix-I.

(b) Adjustment of grating for normal incidence:

(iii) Mount the grating on the prism table in such a way that the rays falling on the grating is normal to it.

(iv) For this the collimator and the telescope are brought in one line and the image of the slit is focused on the telescope. The corresponding vernier scale readings are noted.

(v) Rotate the telescope at angle 90^0 and after that mounts the grating on the prism table so that the reflected image of the slit is seen on the telescope (Fig-4). Take the readings.

(vi) Turn the prism table through an angle 45^0 or 135^0 from this position away from the collimator. This is the position where the grating is normal to the incidence ray.

(c) Measurement of diffraction angles:

(vii) In the position of normal incidence the image of the slit is adjusted to have a bright and sharp image.

(viii) By rotating the telescope, observe the primary maxima for different line spectrum (in different colours) on both the sides of the central point in different orders (Fig-5).

(ix) Rotate the telescope to extreme right end and select one of the prominent line spectra of the 2nd order spectrum and focus it so that the line is seen on the vertical cross wire of the eye piece. Then note the readings of the two vernier scales. In the same order note down the reading of the vernier for other lines (i.e. colour).

(x) Rotate the telescope from extreme right to left and note down the readings of the two verniers for all the spectral lines starting from 2nd order to -2nd order i.e. from extreme right to extreme left.

(xi) The difference of these two pair of readings which is twice of the angle of dispersion is calculated. This reading is repeated twice to get the mean value of the angle of dispersion θ.

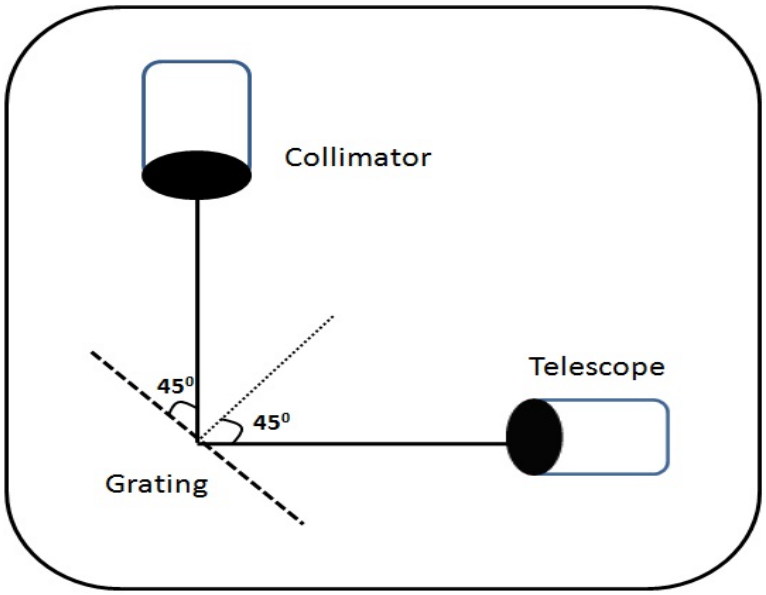

Fig-4: Normal incidence of grating

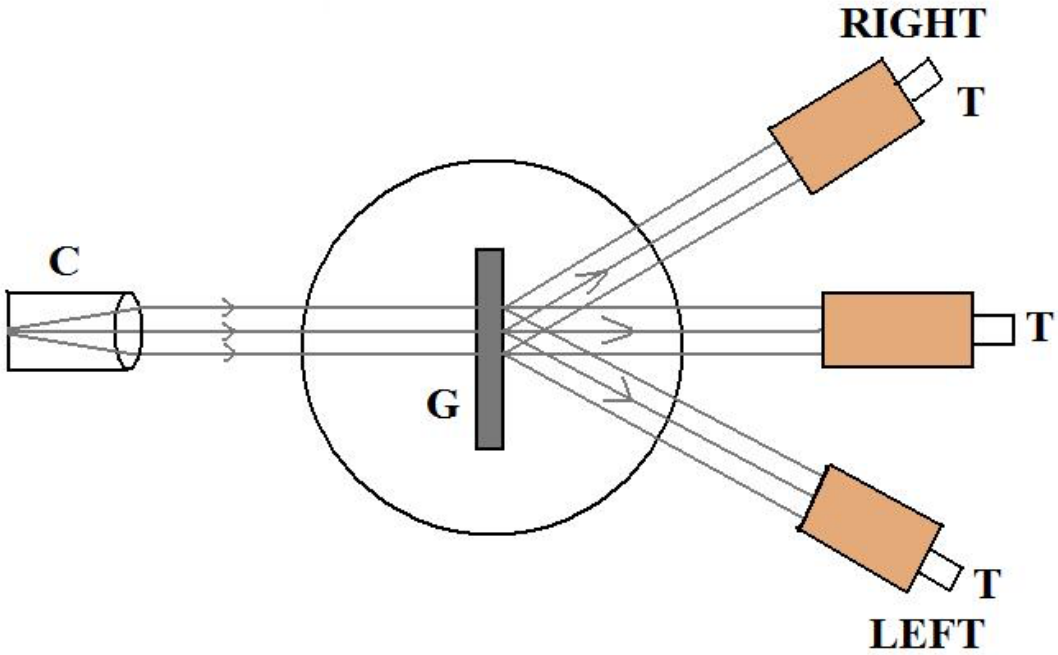

Fig-5: Spectrum by diffraction grating

Observations:

Table-1: Determination of the angle of diffraction for Violet lines.

Colour	V Verner	Readings for the diffracted images with the telescope at the								Difference (2θ)	Mean (2θ)= (a+b)/2	Angle of diffraction (θ)
		Left position				Right position						
		MSR (M)	VSR (V)	TOTAL (M+V)	MEAN	MSR (M)	VSR (V)	TOTAL M+V	MEAN			
VIOLET	1st	162	20/60	162.333	a_1= 162.33	192.5	10/60	192.667	a_2 = 192.66	a =30.33	30.375	15.1875
	2nd	342	8/60	342.133	b_1 = 342.13	12.5	3/60	12.55	b_2 = 12.55	b = 30.42		
	1st	146.5	4/60	146.566	a_1 = 146.56	208.5	13/60	208.716	a_2 = 208.71	a = 62.156	62.086	31.043
	2nd	326.5	6/60	326.6	b_1 = 326.6	28.5	7/60	28.616	b_2 = 28.616	b = 62.016		

Table-2: Determination of the angle of diffraction for Blue lines.

Colour	V (Verner)	Readings for the diffracted images with the telescope at the								Difference (2θ)	Mean (2θ)= (a+b)/2	Angle of diffraction (θ)
		Left position				Right position						
		MSR (M)	VSR (V)	TOTAL (M+V)	MEAN	MSR (M)	VSR (V)	TOTAL M+V	MEAN			
BLUE	1st	159.5	9/60	159.65	159.65	194	12/60	194.2	194.2	a= 34.55	34.495	17.2475
	2nd	339.5	22/60	339.866	339.866	14	18/60	14.3	14.3	b= 34.44		

	1st	143.5	24/60	143.9	143.9	211.5	5/60	211.583	211.583	a= 67.683	67.808	33.904
	2nd	323.5	12/60	323.7	323.7	31.5	8/60	31.633	31.633	b= 67.933		

Table-3: Determination of the angle of diffraction for Yellow lines.

Colour	V (Verner)	Readings for the diffracted images with the telescope at the								Difference (2θ)	Mean (2θ)= (a+b)/2	Angle of diffraction (θ)
		Left position				Right position						
		MSR (M)	VSR (V)	TOTAL (M+V)	MEAN	MSR (M)	VSR (V)	TOTAL M+V	MEAN			
YELLOW	1st	157.5	12/60	157.7	157.7	202	4/60	202.066	202.066	a= 44.366	44.341	22.1705
	2nd	337.5	18/60	337.8	337.8	22	7/60	22.116	22.116	b=44.316		
	1st	135.5	9/60	135.65	135.65	229	19/60	229.316	229.316	a= 93.666	93.541	46.7705
	2nd	315.5	15/60	315.75	315.75	49	10/60	49.166	49.166	b= 93.416		

As per the formula $\lambda=\frac{\sin\theta}{nN}$, *the value of the wave length for different colour can be determined by putting the value of the corresponding θ and n.*

Table-5: Determination of the dispersive power and resolving power of the grating

Colour of the line	No of order	Angle of Diffraction (θ)	No. of lines per cm of the grating	Resolving power of the grating	Dispersive power of the grating
Violet	1 2	15.1875 31.043	6000	15240 30480	15791.540 35575.038
Blue	1 2	17.2475 33.904	6000	15240 30480	15957.565 36724.0709
Yellow	1 2	22.1705 46.7705	6000	15240 30480	16456.7281 44501.39554

Colour of Light	Grating constant (a+b)=1/N	Angle of diffraction(1st order)	Angle of diffraction(2nd order)	Wavelength λ (a+b) sinθ (1st order)(nm)	Wavelength λ (a+b) sinθ (2nd order)(nm)	λ mean (nm)
Violet	$\frac{1}{6000}$	*15.1875*	*31.043*	*436.701*	*429.734*	*433.217*
Blue	$\frac{1}{6000}$	*17.2475*	*33.904*	*494.124*	*464.835*	*479.479*
Yellow	$\frac{1}{6000}$	*22.1705*	*46.7705*	*628.94*	*607.18*	*618.06*

Percentage of Error

We have $\lambda=\frac{sin\theta}{nN}$.

The percent of error $\frac{d\lambda}{\lambda}=100\ \frac{cos\theta d\theta}{sin\theta}=100\ \frac{d\theta}{tan\theta}$

dθ can be taken as the vernier constant and θ is the concerned angle of dispersion.

In case of a spectrometer, there is a circular scale marked with 0 to 360^0. Here each single division is one degree which is again divided into 2 divisions of half of a degree each. So the smallest division of the scale is 0.5 degree. Further,

60 VSD = 59 MSD, so 1 VSD = 59/60 MSD

So least count =

1 MSD – 1 VSD

$= (1- 59/60)$ MSD $= (1/60 \times 0.5)^o = 1.4544 \times 10^{-4}$ radians

Case I: For Violet,

$\boldsymbol{\lambda}_1$(Wavelength First Order) = 436.701nm

$\boldsymbol{\lambda}_2$ (Wavelength Second Order) = 429.734nm

$\text{Error}_{\text{1st Order}} = \frac{d\lambda_1}{\lambda_1} = \frac{d\theta}{tan\theta} = \frac{1.4544 \times 10^{-4}}{\tan(15.1875)} = 5.3577 \times 10^{-4}$

% $\text{Error}_{\text{1st Order}} = 0.054\%$

So, $d\lambda_1 = 5.3577 \times 10^{-4} \times \lambda_1 = 0.23397$

Similarly,

$\text{Error}_{\text{2nd Order}} = \frac{d\lambda_2}{\lambda_2} = \frac{d\theta}{tan\theta} = \frac{1.4544 \times 10^{-4}}{\tan(31.043)} = 2.4164 \times 10^{-4}$

% $\text{Error}_{\text{2nd Order}} = 0.024\%$

So, $d\lambda_2 = 2.4164 \times 10^{-4} \times \lambda_2 = 0.10384$

$\text{Error}_{\text{Violet}}$

$= \frac{d\lambda}{\lambda} = \frac{d(\frac{\lambda_1+\lambda_2}{2})}{\frac{\lambda_1+\lambda_2}{2}} = \frac{d\lambda_1+d\lambda_2}{\lambda_1+\lambda_2} = \frac{0.23397+0.10384}{436.701+429.734} = 0.00039$

% $\text{Error}_{\text{Violet}}$ = 0.039%

Case II: For Blue,

$\boldsymbol{\lambda}_1$(Wavelength First Order) = 494.124nm

$\boldsymbol{\lambda}_2$ (Wavelength Second Order) = 464.835nm

$\text{Error}_{\text{1st Order}} = \frac{d\lambda_1}{\lambda_1} = \frac{d\theta}{tan\theta} = \frac{1.4544 \times 10^{-4}}{\tan(17.2475)} = 4.6846 \times 10^{-4}$

% $\text{Error}_{\text{1st Order}} = 0.047\%$

So, $d\lambda_1 = 4.6846 \times 10^{-4} \times \lambda_1 = 0.23148$

Similarly,

$\text{Error}_{\text{2nd Order}} = \frac{d\lambda_2}{\lambda_2} = \frac{d\theta}{tan\theta} = \frac{1.4544 \times 10^{-4}}{\tan(33.904)} = 2.1640 \times 10^{-4}$

% $\text{Error}_{\text{2nd Order}} = 0.022\%$

So, $d\lambda_2 = 2.1640 \times 10^{-4} \times \lambda_2 = 0.10059$

$\text{Error}_{\text{Blue}}$

$= \frac{d\lambda}{\lambda} = \frac{d(\frac{\lambda_1+\lambda_2}{2})}{\frac{\lambda_1+\lambda_2}{2}} = \frac{d\lambda_1+d\lambda_2}{\lambda_1+\lambda_2} = \frac{0.23148+0.10059}{494.124+464.835} = 0.00035$

% $\text{Error}_{\text{Blue}}$ = 0.035%

Case III: For Yellow,

$\boldsymbol{\lambda}_1$(Wavelength First Order) = 628.940nm

λ_2 (Wavelength Second Order) = 607.180nm

$\text{Error}_{1\text{st Order}} = \frac{d\lambda_1}{\lambda_1} = \frac{d\theta}{tan\theta} = \frac{1.4544 \times 10^{-4}}{\tan(22.1705)} = 3.5691 \times 10^{-4}$

% $\text{Error}_{1\text{st Order}} = 0.036\%$

So, $d\lambda_1 = 3.5691 \times 10^{-4} \times \lambda_1 = 0.22447$

Similarly,

$\text{Error}_{2\text{nd Order}} = \frac{d\lambda_2}{\lambda_2} = \frac{d\theta}{tan\theta} = \frac{1.4544 \times 10^{-4}}{\tan(46.7705)} = 1.3672 \times 10^{-4}$

% $\text{Error}_{2\text{nd Order}} = 0.014\%$

So, $d\lambda_2 = 1.3672 \times 10^{-4} \times \lambda_2 = 0.08301$

$\text{Error}_{\text{Yellow}}$

$= \frac{d\lambda}{\lambda} = \frac{d(\frac{\lambda_1+\lambda_2}{2})}{\frac{\lambda_1+\lambda_2}{2}} = \frac{d\lambda_1+d\lambda_2}{\lambda_1+\lambda_2} = \frac{0.22447+0.08301}{628.940+607.180} = 0.00025$

% $\text{Error}_{\text{Yellow}} = 0.025\%$

Precaution

(i) The spectrometer should be levelled properly.

(ii) The spectral lines should be adjusted at the vertical cross wire of the telescope

(iii) Care must be taken so that while taking observation the source, the prism and the collimator should not be displaced.

(iv) The grating should be handled from the edge

Considerations

- What is a plane transmission diffraction grating and how diffraction occurs?

  Ans: The plane transmission grating is a plane sheet of transparent material on which opaque rulings are made so that it acts as a diffraction grating with multiple slits. The resulting diffraction pattern is due to multi-slit diffraction.

- How grating is made and what is a grating element?

Ans: It can be made up of transparent material on which opaque rulings are made with a fine diamond pointer or laser interference lithography. The combined width of a ruling(opaque part) and a slit is called grating element .

- What is the effect of increase in number of lines of the grating on the diffraction pattern?

Ans: When we increase the number of lines in a diffraction grating, effectively the grating element decreases and the resolving power goes up. The maxima corresponding to different closely spaced wavelengths become spread more far apart

- Define dispersive power and resolving power of an optical instrument?

Ans: The resolving power of an optical instrument is defined as the reciprocal of the minimum angular separation that allows to just resolve two closely spaced images. The dispersive power is the ability of an instrument to separate out light having closely spaced wavelengths into its constituents. It is defined as $d\theta/d\lambda$ where $d\lambda$ is the difference between 2 closely spaced wavelengths and $d\theta$ is the angular separation between the directions into which they are scattered.

- How does angular dispersive power of the grating depends upon
    1. Order of spectrum(n): directly proportional
    2. Grating element: inversely proportional
    3. Wavelength (λ): independent
- What will happen if the rulings of the grating are not parallel and the distance between two consecutive rulings is not a constant?

Ans: In this case also diffraction occurs but an irregular fringe pattern will be formed. The calculations and representations may be quite complicated.

- What are the uses of grating?

Ans: Gratings are widely used as dispersive element and finds notable applications in spectrometers, monochromators, astronomical spectroscopy, and study of fluids because of their sensitive dependence on refractive index of the material.

Conclusions

(i) Diffraction gratings are particularly ingenious optical instruments. It uses wave nature (interference) of light to separate out light into its constituent components. The do not require moving parts, are compact and simple. They are widely

used to analyse the spectra of unknown sources like emission/absorption spectra of atoms, light from distant galaxies to name a few.

(ii) Essentially in this experiment our eyes operate the vernier scale and the readings are as correct as our eyes are in accurately identifying the particular wavelengths. For 2 closely spaced wavelengths our eyes might not be able to tell them apart in lower order maxima. Barring that, gratings are very accurate instruments.

(iii) Reflective/ transmission diffraction gratings are widely used in modern science. They can be used to analyse unknown spectra and filter out or select a particular wavelength form a spectra :something that cannot be done properly by other dispersive elements like prisms(due to low dispersive power, resolving power). They find various applications in spectrometers, astronomical spectroscopy, monochromators, study of fluids, measuring refractive indices and many more .